SANKUR KUNDU

ML Systems Engineer | LLM Training, Inference & GPU Kernels
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SUMMARY

ML systems engineer (B.Tech CSE, 2026) and sole author of four manuscripts. Six fixes merged in 11 days into **Apple MLX**, **Meta ExecuTorch**, **Hugging Face**, **Oxford's PyZX** and **JaxMARL**. I build ML systems from scratch and benchmark them against the leaders: kernels at **cuBLAS** parity, an LLM engine matching **vLLM** token for token, and a tokenizer beating **GPT-4o's**.

OPEN-SOURCE IMPACT: SIX FIXES MERGED IN 11 DAYS

- **Apple MLX:** fixed a data race in the distributed training backend's socket threads (ThreadSanitizer: 2–4 races per run → **0**); merged by the MLX maintainer. #4737
- **Meta ExecuTorch:** closed an out-of-bounds read in the 2- and 4-bit quantized embedding kernels of on-device LLMs; approved by two Meta maintainers: "LGTM! Thanks for the fix!" #23677
- **JaxMARL** (NeurIPS 2024; Oxford, UCL & Imperial): fixed RL rollouts that zeroed every new episode's first observation; merged by the lead author: "Good find!!" #206
- **Oxford PyZX:** fixed a circuit-optimiser rewrite that broke **25 of 100** random Clifford circuits; merged by its co-creator: "This was a regression". #563
- **Hugging Face OpenEnv:** fixed reward rubrics for RL training that ran hooks out of order and scored async judges as coroutines; "177 rubric tests pass". #1441, #1442
- **In review:** Google JAX (assigned to JAX co-author Matthew Johnson), NVIDIA, vLLM, PyTorch, MIT, Carnegie Mellon, ETH Zurich.

RECOGNITION

- **NASA CARA** analysts are reviewing my reproduction of its collision results (**53/53** cases); **Entrepreneur First** (UK) scouted me from public work (evidence).

RESEARCH

- Sole author: **"Removing the Token Tax from an Existing Model"**, **"The Sampler Was Not the Policy"**, **"Reproducing NASA CARA's Monte Carlo Probability of Collision"** (arXiv, forthcoming).

PROJECTS

ratchet — PyTorch, C++/CUDA: RL post-training (GRPO) from scratch · study page

- Lifted Qwen2.5-0.5B on GSM8K from **48.1% to 52.5%** in 100 steps on 2× T4, scoring all 1,319 test problems.
- Cut weight sync **369×** with in-place GPU updates, and made steps **1.9×** faster by overlapping generation with training.
- Found fp16 weight sync silently erasing most updates, so the sampler ran a policy the trainer never had (gap **5.2 → 0.04** nats).

tessel — Rust: a tile language and compiler for GPU and TPU kernels · study page

- Wrote a Triton-style compiler: tile IR, tensor-core CUDA (mma.sync, ldmatrix, swizzled shared memory, pipelining), auto-tuning, Metal and TPU backends.
- Matched **cuBLAS** on GEMM (**1.00–1.04×**) and beat PyTorch SDPA on causal attention (**1.3–1.9×**) on a T4, checked against fp64.
- Built an LLM engine on these kernels alone that matches **vLLM token for token at 1.05× its throughput** (TinyLlama-1.1B).

lipi — PyTorch: cutting the token tax on Indian languages · study page

- Built a tokenizer that needs fewer tokens than **GPT-4o's and Gemini's in 23 of 24** Indian languages (on Hugging Face).
- Extended Qwen2.5-0.5B with it: Odia takes **8.6×** fewer tokens and, in equal GPU time, is predicted **20% better**.

samebit — C, PyTorch, MLX: integer-only LLM inference with the same bits on every chip

- Made **five independent backends** (C/OpenMP, NumPy, PyTorch, **Apple MLX**, reference) give **one SHA-256 digest over 153 tensors** on x86-64 and ARM64.

More projects

- **relay**: fault-tolerant LLM serving on custom kernels, prefill **8.7×** faster, **384/384** requests identical through 60 kills.
- **syndrome**: GPU quantum decoder bit-identical to **IBM's**, **28.7×** faster per shot; **kessler**: matches **NASA CARA** on 53/53.
- **switchyard**: MoE serving **2.2×** Hugging Face; **kiln**: beats torch.compile on CPU; **crucible**: **0 of 9** reward hacks pass.

EXPERIENCE

Gigzen — Co-founder & Founding Engineer **Apr 2026 – Present**

Early-stage startup (name approved by India's MCA; incorporation in progress) · Sole engineer

- Built Waggle, three Android delivery apps on one PostgreSQL backend: **53 tables, 90 RLS policies, 154 SQL functions**; **150 automated attacks**, all blocked.
- Moved dispatch, UPI reconciliation and GST billing into the database, so no client can bypass them.

SKILLS

ML systems: PyTorch, JAX, MLX; GRPO, DDP, ZeRO/FSDP, MoE expert parallelism, paged KV cache, int8 quantization
Languages & GPU: Python, C++20, CUDA (tensor cores), Rust, C, Swift, SQL; Triton-style kernels, Metal, Pallas (TPU)

EDUCATION

B.Tech, Computer Science and Engineering (AI & ML) — IEM Kolkata **Sep 2022 – Apr 2026**